What's Needed

Now that we know why we are learning to write code and generally what this platform offers, we are going to need a little bit of hardware to complete the projects discussed in the rest of this book. Even though we will focus our discussions primarily on programming the Arduino, the hardware is what makes programming the Arduino so much fun. The best place to get started with prototyping electronic circuits with the Arduino is through one of the many starter kits available from various retailers for anywhere from $40 to $100 USD or about €30 to €70 EUR. SparkFun Electronics (`www.sparkfun.com`) offers the Starter Kit and Inventor's Kit; Adafruit Industries (`www.adafruit.com`) the Budget Pack or Starter Pack; *MAKE Magazine*'s Maker SHED (`www.makershed.com`) with either a Getting Started Kit or the Mintronics Survival Pack; or the Arduino Store's (`http://store.arduino.cc`) Workshop Kit. Figure 1-7 shows what this assembled kit might look like.



Figure 1-7. *Getting started hardware kit*

While each of these kits will have more or less what you need, Table 1-1 provides a simple shopping list of some of the specific hardware that we will use in the following pages, which you might want to purchase individually or at least check whether or not the kit you are considering contains these items. Prices are only approximate and are listed just to give you an idea on the costs.

Keep in mind that this is just a general list of recommended parts; feel free to purchase something different—as long as it's somewhat similar to what's listed—and to order from vendors of your choice. For example, rather than using an Arduino Uno, the standard Arduino interface board, maybe you're interested in wearables or e-textiles and would like to use a LilyPad Arduino instead. That's fine, just be sure to read up on the differences between the recommended hardware and your own on the Arduino Hardware page at `http://arduino.cc/en/Main/Hardware`. Likewise, many of the items on the list, such as resistors, are kind of like horseshoes in that close enough will usually get the job done. For a more detailed list that includes all the parts used in this book, refer to the expanded hardware list in the appendix at the end of this book.